

RFI Report for August 2024

Executive summary

A few datasets have been sampled during the month of August in order to investigate the RFI status in the UHF-band. Based on the few samples, overall, the cross correlation products, which are mainly the most important for science, are reasonably cleaner than the autocorrelation. The investigation is extended by considering all observations conducted in the month of August but only cross correlations were considered.

Overall, RFI levels for August remain generally low, exhibiting moderate variability throughout the day. Notably, a significant spike occurs at 17:00 UTC, with levels exceeding 80%, while otherwise, fluctuations are steadily and lower. The ECDF plot indicates that most RFI values fall between 10 and 20, with higher RFI intensities being rare, suggesting that severe RFI events are infrequent but impactful when occurs and these are due to known RFI sources and therefore not construction based.

1. Introduction

The purpose of this document is to report the current status of RFI on-site. Given the construction work that has been going on over the last couple of months, it is therefore imperative to monitor construction related activities on site that could potentially generate RFI signals which, in turn, contaminate the science observations. Preferably these reports are to be released on a weekly basis (or every two weeks).

2. Method

The flags data, specifically *ingest_flags*, are considered in these analyses, as they can be used as proxies of RFI frequency occupancy, i.e. RFI emission as a function of frequency channels. The strategy is to select one to three observations per week and report the status of the RFI contamination on site based on the flags of each dataset (of each observation). For each observation

- Autocorrelation flags of a few *scans* corresponding to the *gain calibrator* are selected. This is due to the fact that, at the time of writing, there are some issues with accessing a big chunk of an entire dataset.
- Autocorrelation flags, which are three dimensional arrays, i.e. *time: frequency channels: correlation products*, are averaged over both time and correlation products.
- Autocorrelation flags become a one dimensional array which is plotted in order to check the RFI occupancy in the frequency channels.

The cross correlation flags are also investigated in order to see how much the RFI emission affects the correlation products which are the main interest in an interferometric observation. The procedure for the cross correlation flags is the same as that of autocorrelation ones. To

further go into more detail, the flags corresponding to short and long baselines are split into two parts, and each part is plotted separately. Based on some datasets that were analyzed previously, the U-band observations are the ones that are most affected by the current RFI emissions on site. Hence, the majority of the datasets that will be inspected is in U-band, nevertheless observations in both L and S bands will be looked at too.

In what follows, it is worth noting that the vertical gray shaded areas and lines in all plots denote the known RFI emissions in U-band. For ease of reference, the details of the observations in the next section (Results) are listed in the Table 1 below.

Proposal ID	Schedule Block ID	Capture Block ID	Start time	End time
SCI-20230907-MD-01	20240202-0054	1723102578	2024-08-08 07:35	2024-08-08 12:40
EXT-20240729-SA-01	20240818-0019	1724095875	2024-08-19 19:30	2024-08-19 22:00
SCI-20230907-AG-01	20231218-0002	1724338872	2024-08-22 13:00	2024-08-22 14:02
SCI-20180516-SB-02	20240727-0006	1724968210	2024-08-29 21:30	2024-08-29 06:46

Table 1: Details of each dataset considered for the analyses (Section Results).

3. Results

a. 2024, 08 August

The mean autocorrelation flags of the dataset collected on the 8th of August 2024 is plotted in Figure 1. Some unknown RFI emissions due to some activities on site are noticed at some frequencies. The cross correlation flags for short and long baselines are plotted in Figures 2 and 3 respectively. By comparing the short and long baseline results, it can be inferred that the short baselines antennas are more affected by RFI emissions.

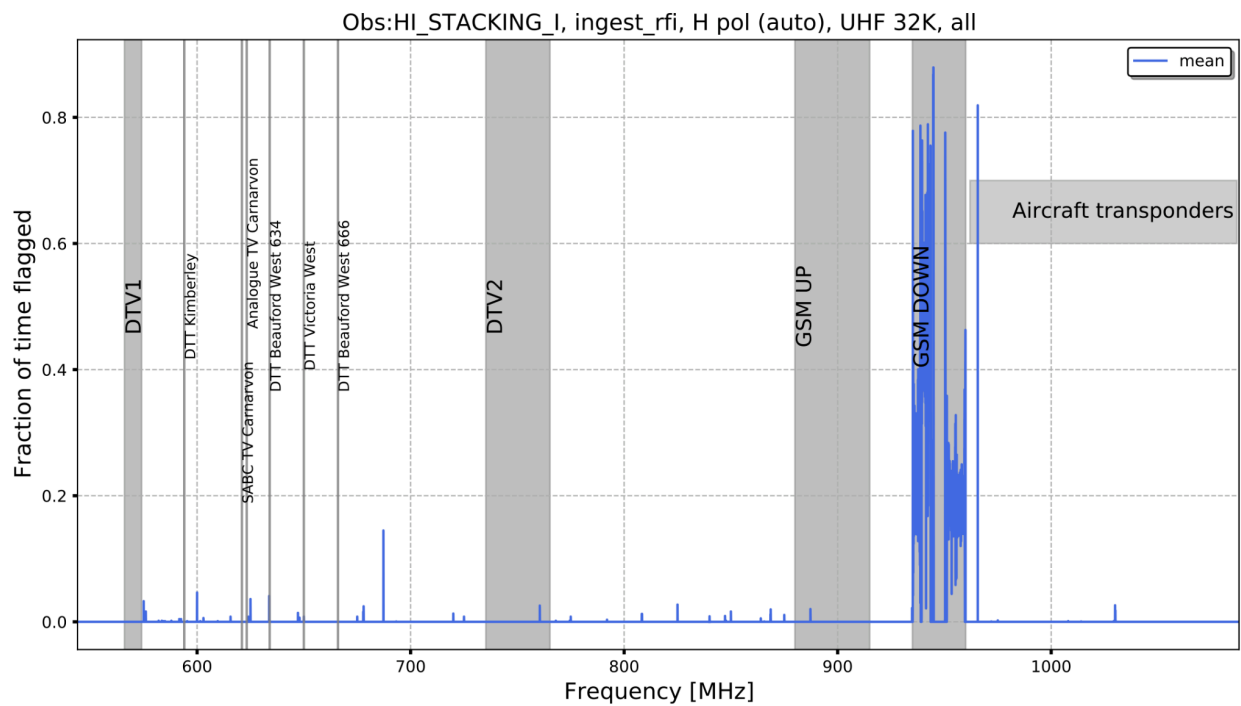


Figure 1: Mean of the autocorrelation flags of SB_ID:20240202-0054 dataset.

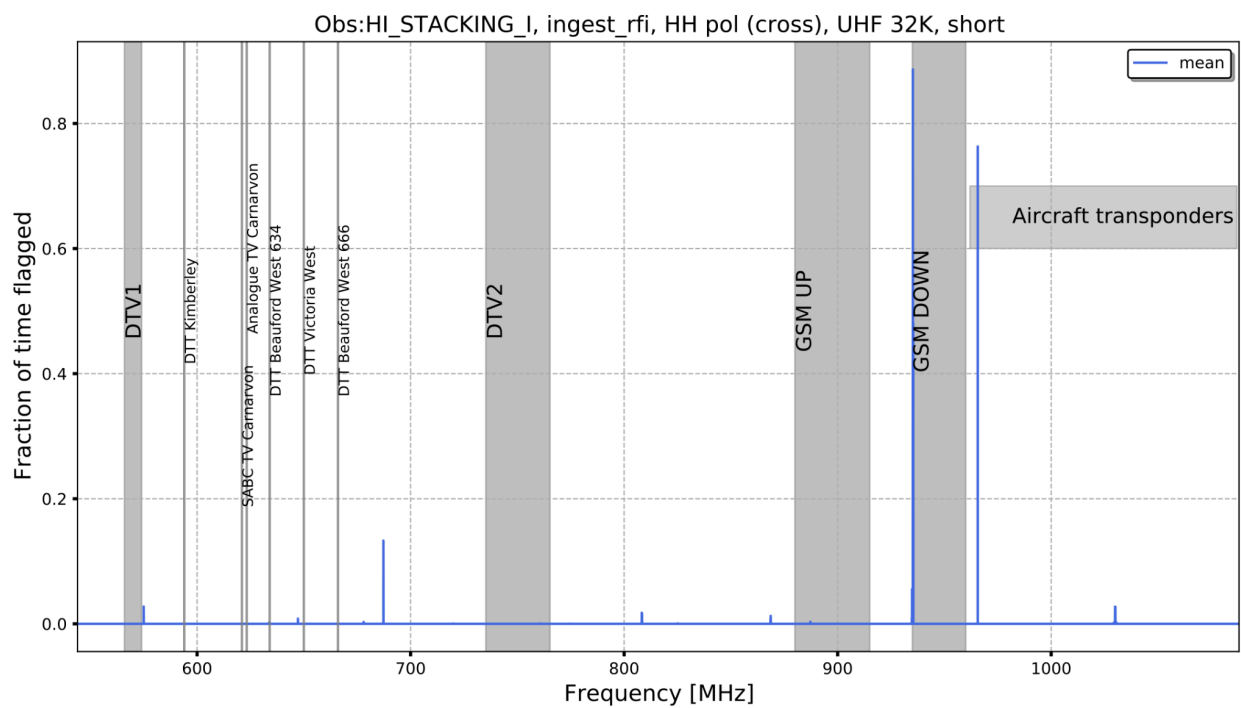


Figure 2: Mean of the cross correlation flags of the short baselines of SB_ID:20240202-0054 dataset.

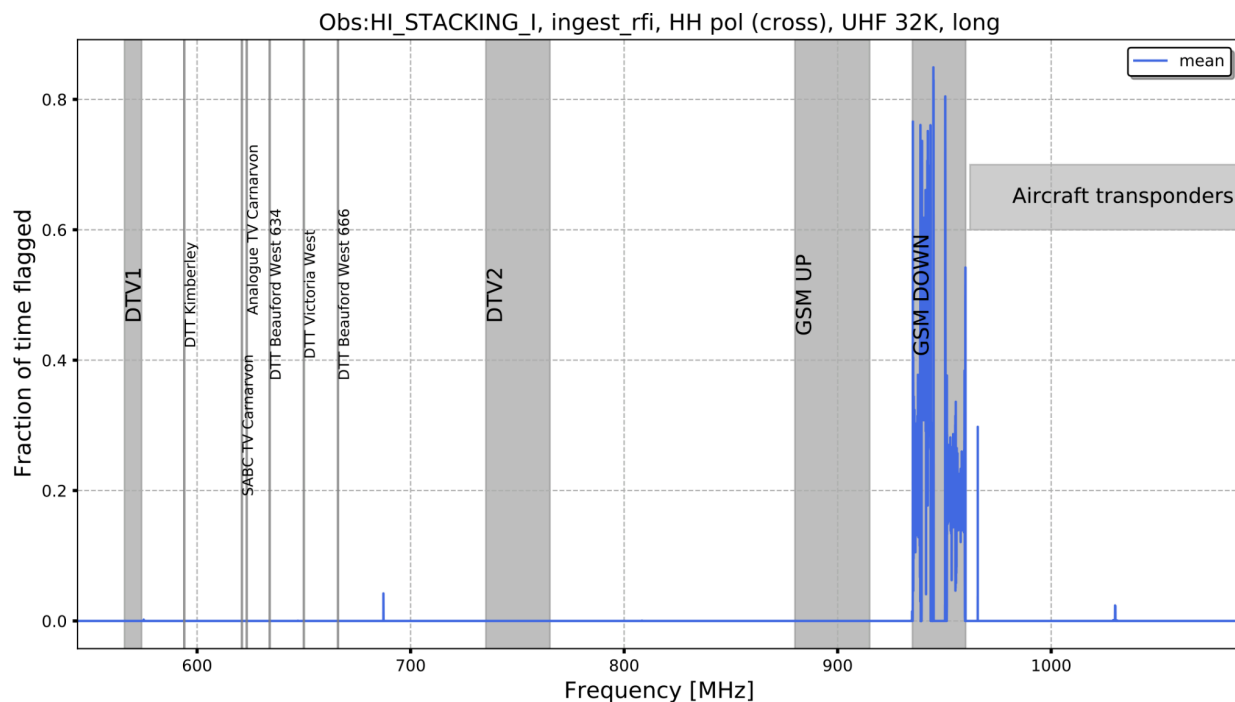


Figure 3: Mean of the cross correlation flags of long baselines of SB_ID:20240202-0054 dataset.

b. 2024, 19 August

The autocorrelation, short and long baselines cross correlations are plotted in Figures 4, 5, 6 respectively. The dataset was collected on the 19th of August 2024. Some unknown RFI emissions, although relatively weak, are seen at some frequencies in Figure 4. Overall the weak RFI emissions are not present in the cross correlation, but a more prominent RFI at around 600 MHz is still noticeable in the cross correlation flags although weaker in the long baselines (Figure 6).

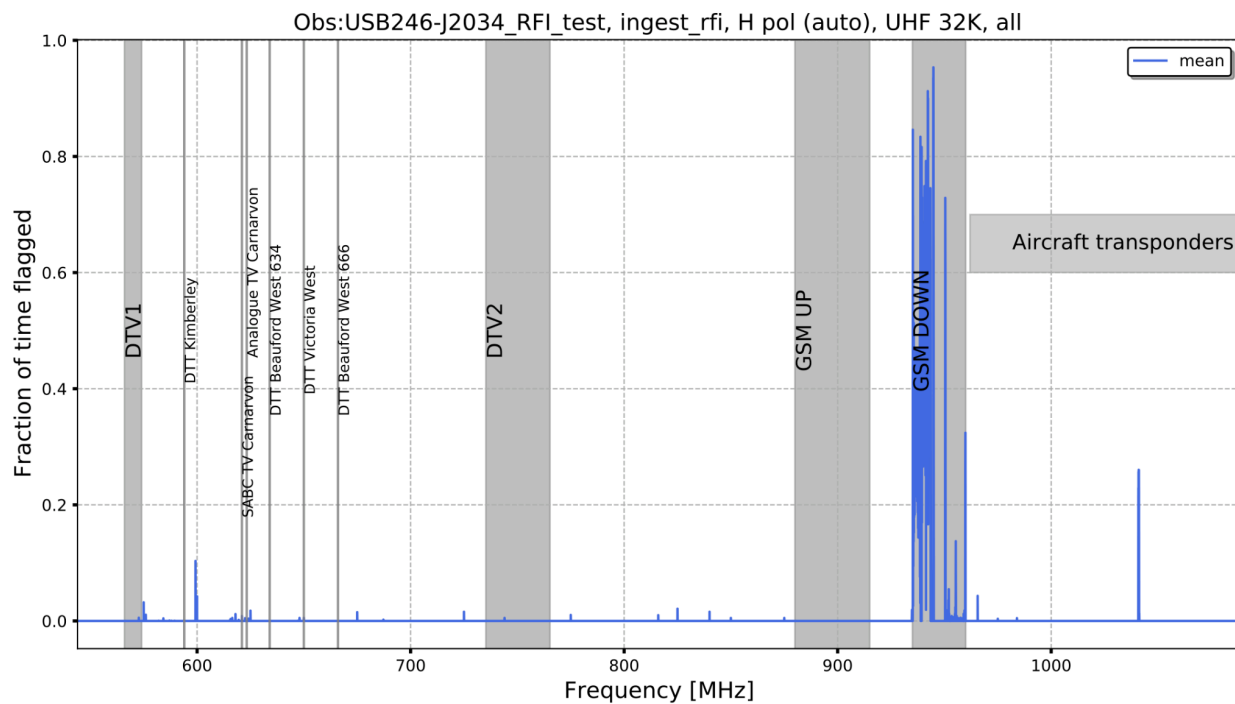


Figure 4: Mean of the autocorrelation flags of SB_ID:20240818-0019 dataset.

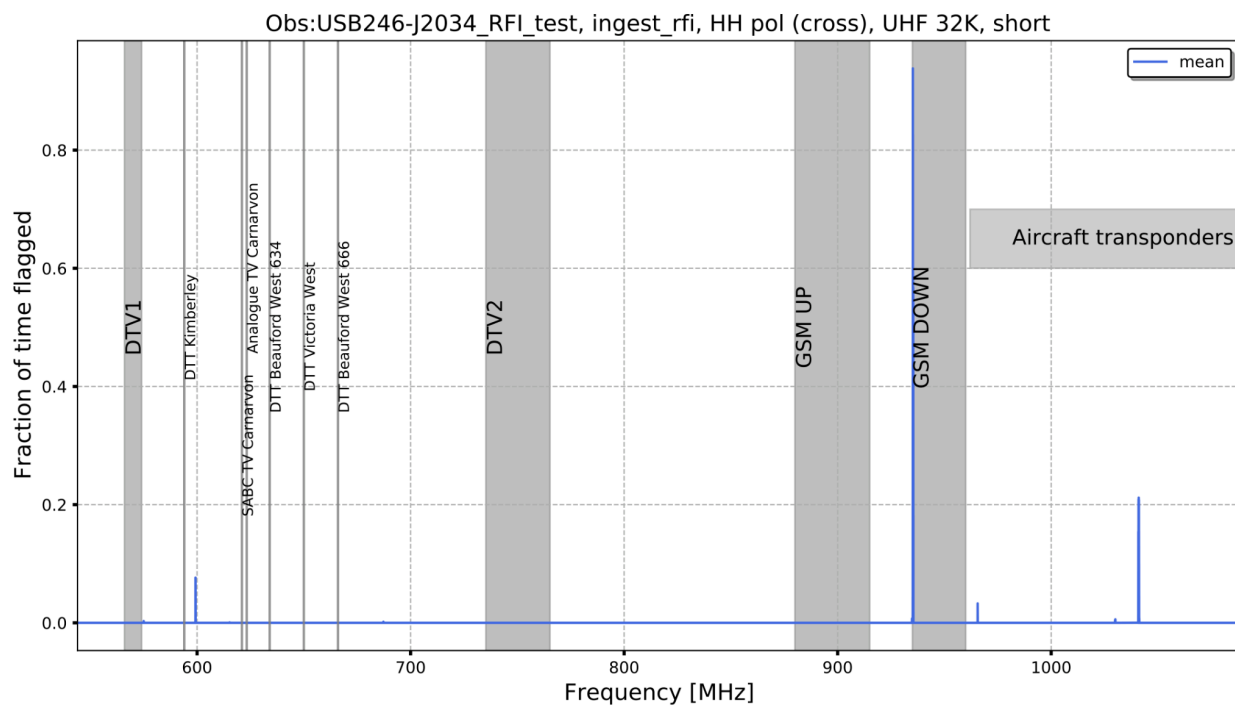


Figure 5: Mean of the cross correlation flags of short baselines of SB_ID:20240818-0019 dataset.

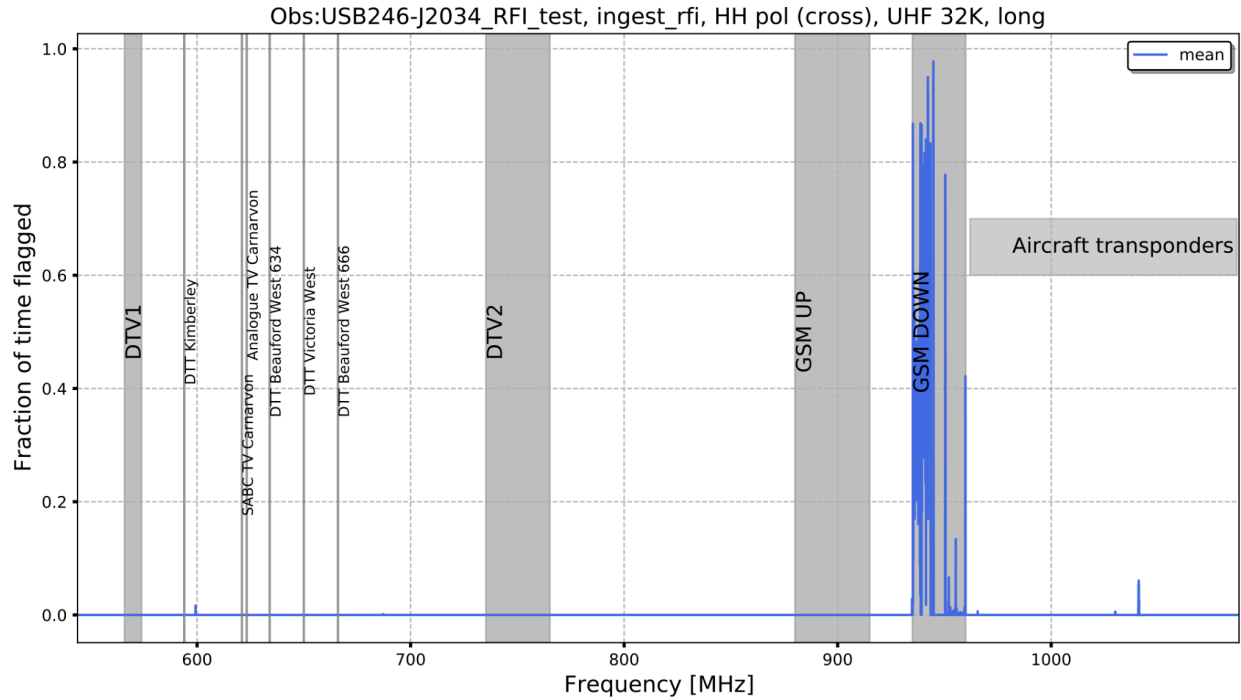


Figure 6: Mean of the cross correlation flags of long baselines of SB_ID:20240818-0019 dataset.

c. 2024, 22 August

The data that is considered in this section was collected on 22nd of August 2024. In Figure 7, the autocorrelation flags are plotted. Unknown RFIs can be clearly seen, although the strength, in other words the fraction of time flagged, is less than 0.1 overall. The cross correlation as shown in Figures 8, 9 is cleaner than the autocorrelation, i.e. the latter is more affected by RFI.

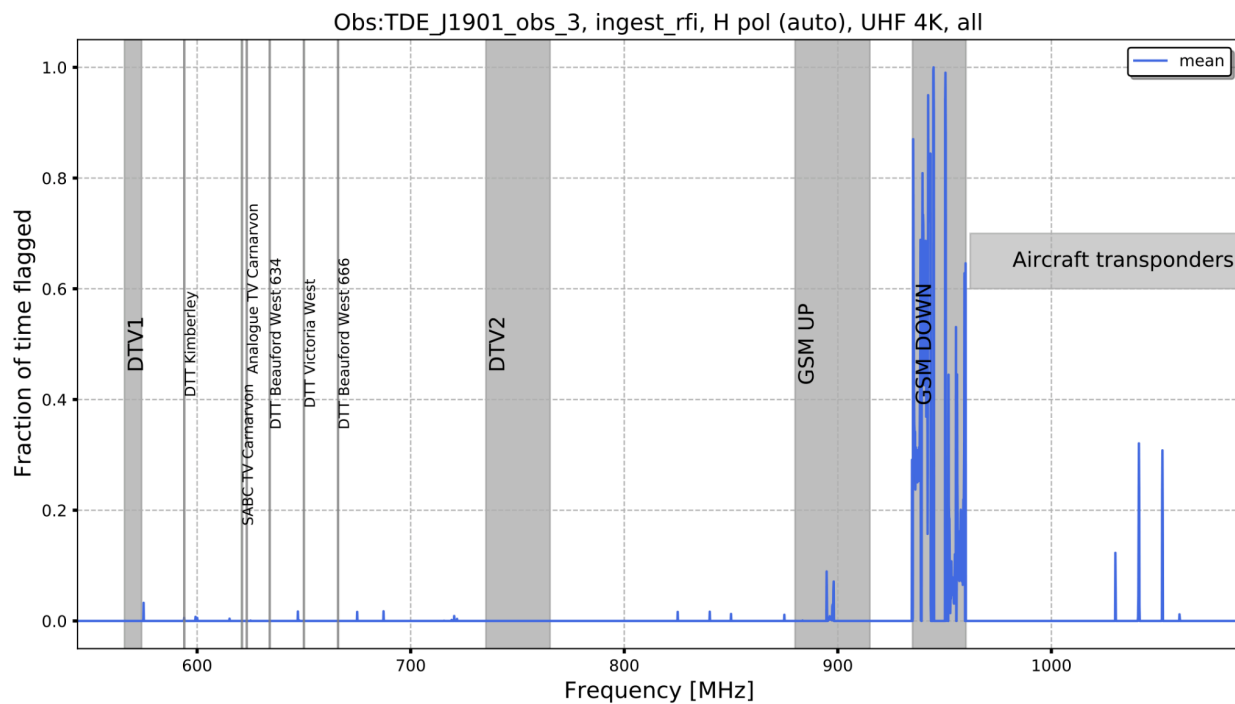


Figure 7: Mean of the autocorrelation flags of SB_ID:20231218-0002 dataset.

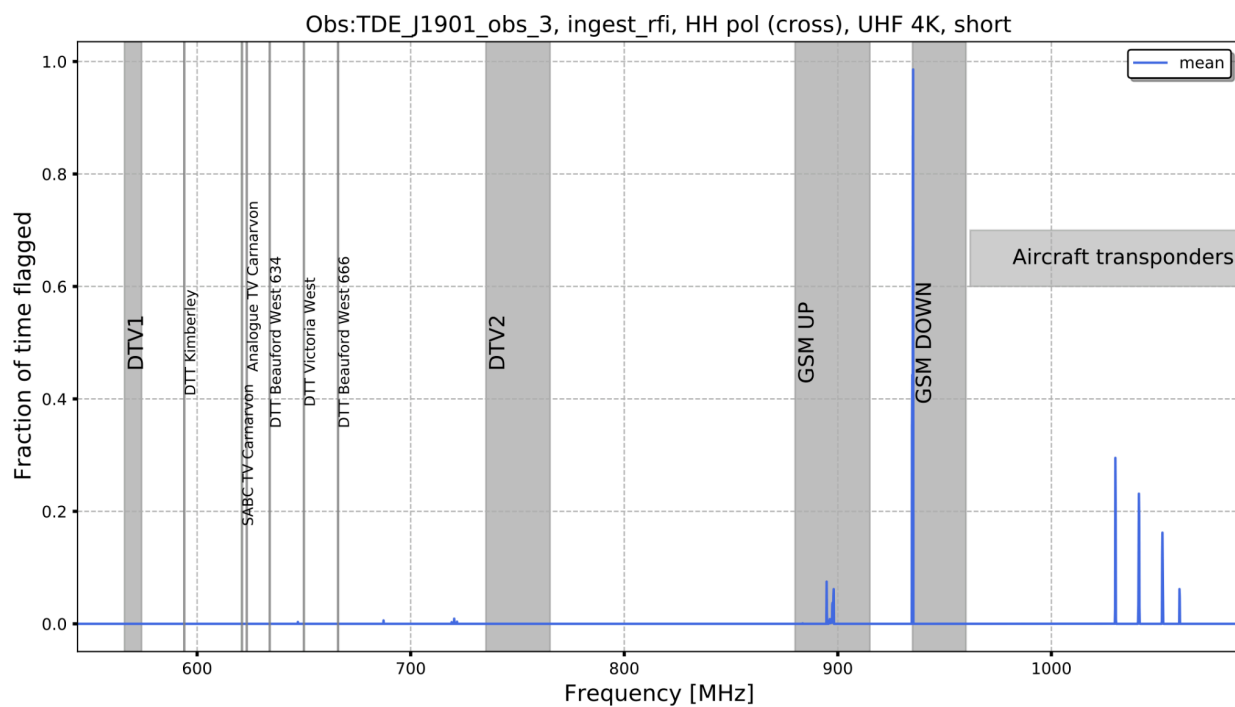


Figure 8: Mean of the cross correlation flags of the short baselines of SB_ID:20231218-0002 dataset.

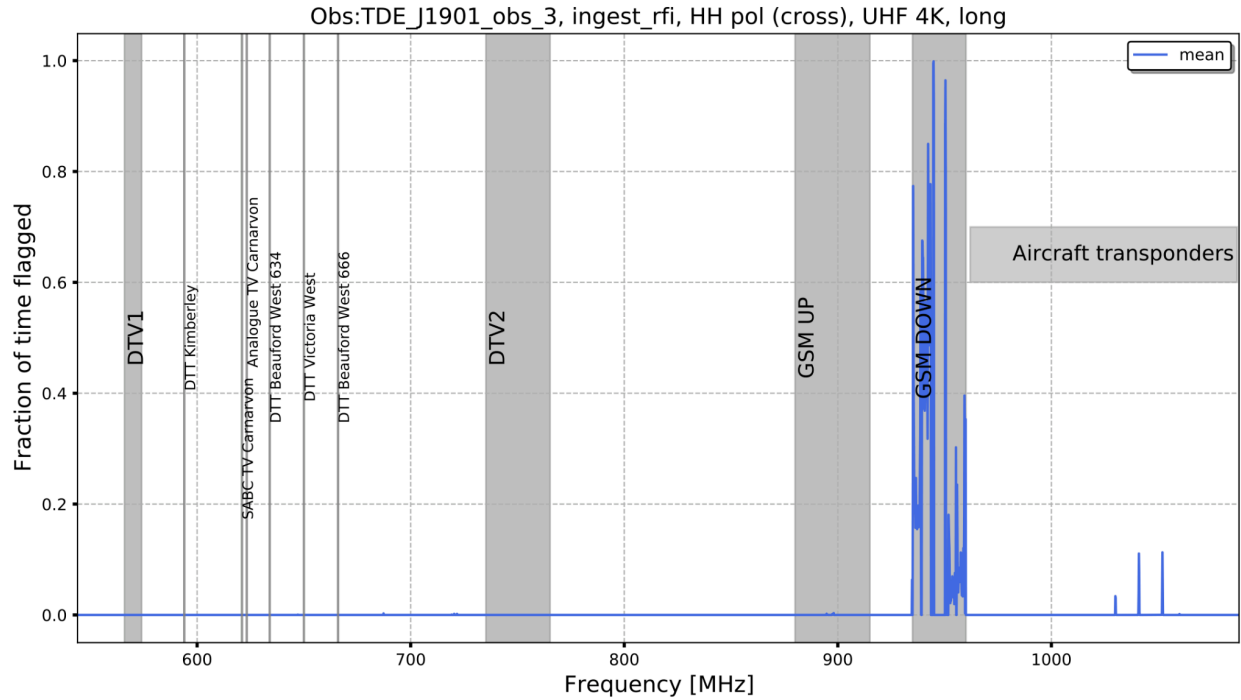


Figure 9: Mean of the cross correlation flags of long baselines of SB_ID:20231218-0002 dataset.

d. 2024, 30 August

The results plotted in Figures 10, 11 and 12 are obtained from a dataset collected on the 30th of August 2024. It is clear from Figures 11 and 12 that the cross correlation is not affected by the relatively weak RFI emissions that are observed in the autocorrelation plots in Figure 10. This might suggest that those interferences seen in the autocorrelation are related to each antenna and do not cross-correlate.

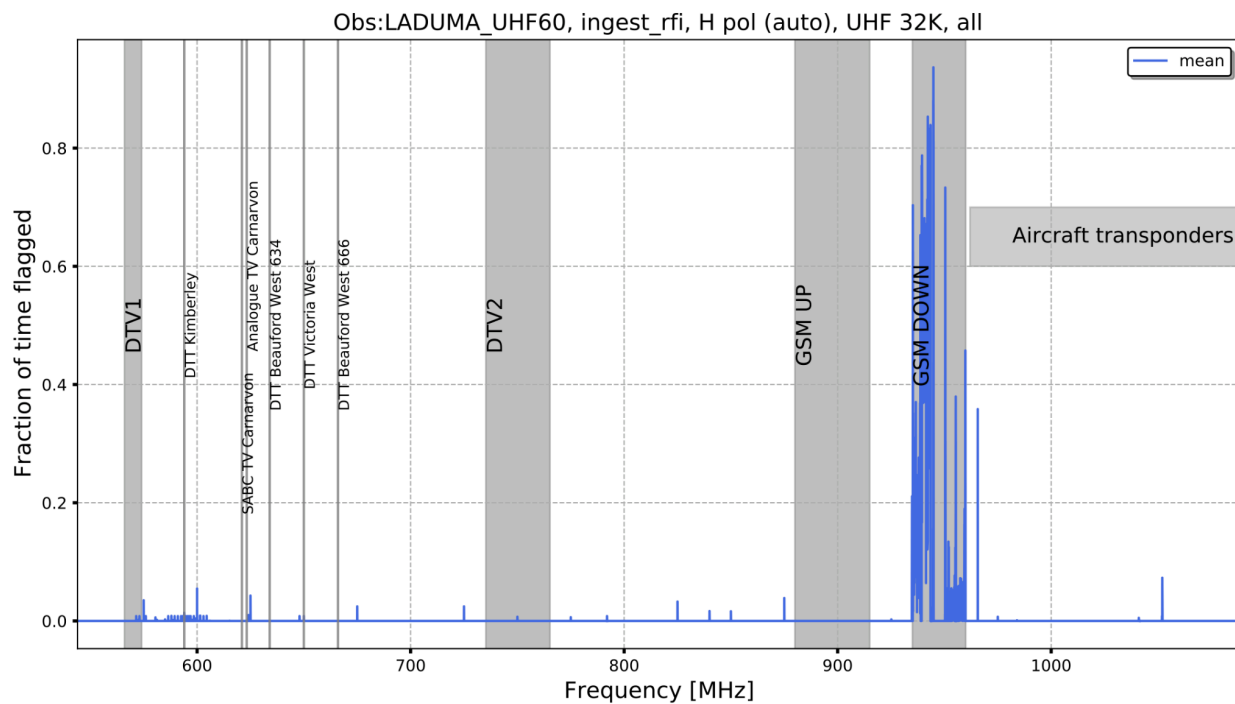


Figure 10: Mean of the autocorrelation flags of SB_ID:20240727-0006 dataset.

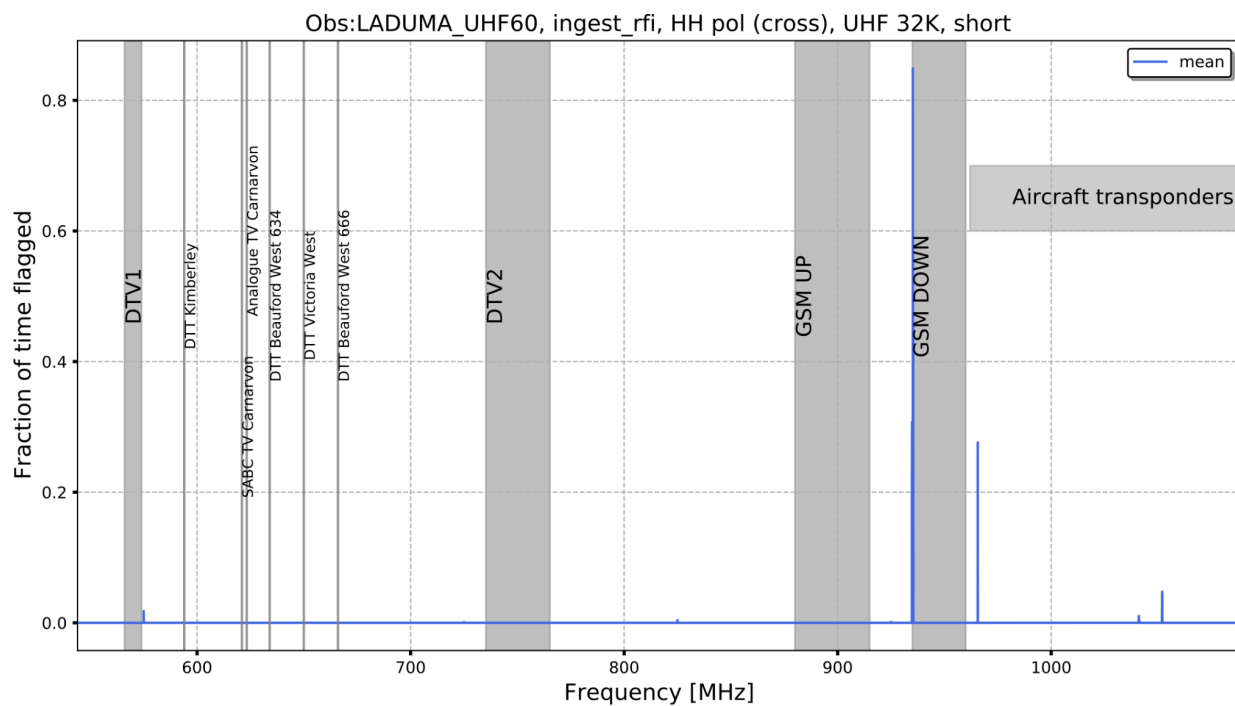


Figure 11: Mean of the cross correlation flags of short baseline of SB_ID:20240727-0006 dataset.

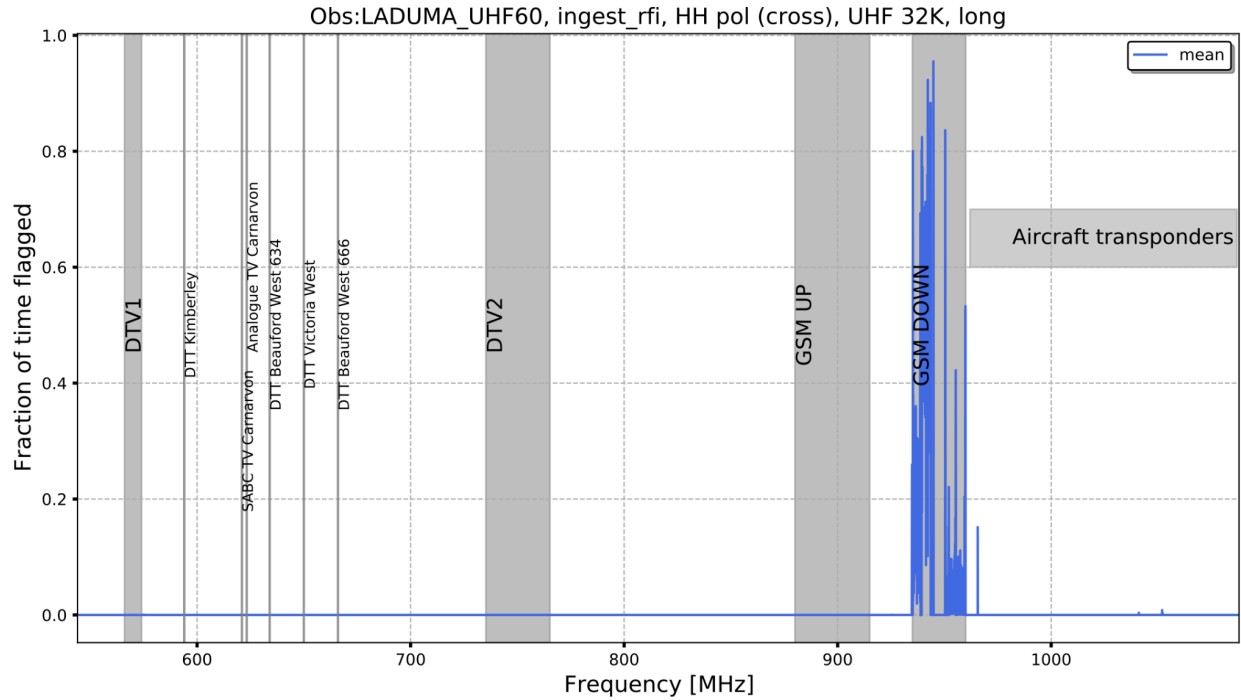


Figure 12: Mean of the cross correlation flags of long baselines of SB_ID:20240727-0006 dataset.

3 . RFI summary for August 2024 at UHF, HH cross polarization

Now we look at the overall RFI for the month of August by concatenating all science observations for the month. Here we select particularly 8s observations, observed at 4k or 32k channel mode. The monitoring and analysis of RFI are presented using the KATHPRFI pipeline [1], highlighting the impact of construction on RFI levels.

KATHPRFI algorithm

The kathprfi essentially processes two numpy arrays, Master and Counter, to update the number of RFI points and observations per voxel which refers to 5-D array of Time, Frequency, Baseline, Elevation and Azimuth. It iterates over chunks of frequency indices, baseline indices, and time indices. For each combination, the algorithm updates the Master array by the corresponding 'Good flags' value and the Counter array by 1. Good flags for a given observation are based on a flag type, e.g. if flag type is calibration RFI then target tags for that observation will be obtained from a complete calibration report. The output of this is an xarray dataset that contains arrays of these two arrays and the dimensions of interest as mentioned above.

Computation of marginal probability

The RFI probability is computed by using two arrays from the output Zarr file obtained from the kathprfi pipeline : 'Master' and 'Counter'. These arrays represent the number of RFI points and the total number of observations, respectively. The function sums these arrays along a specified

dimension to get total counts and then calculates the probability as the ratio of these sums.

RFI amount as a function of time of the day for August 2024 at UHF, HH cross pol

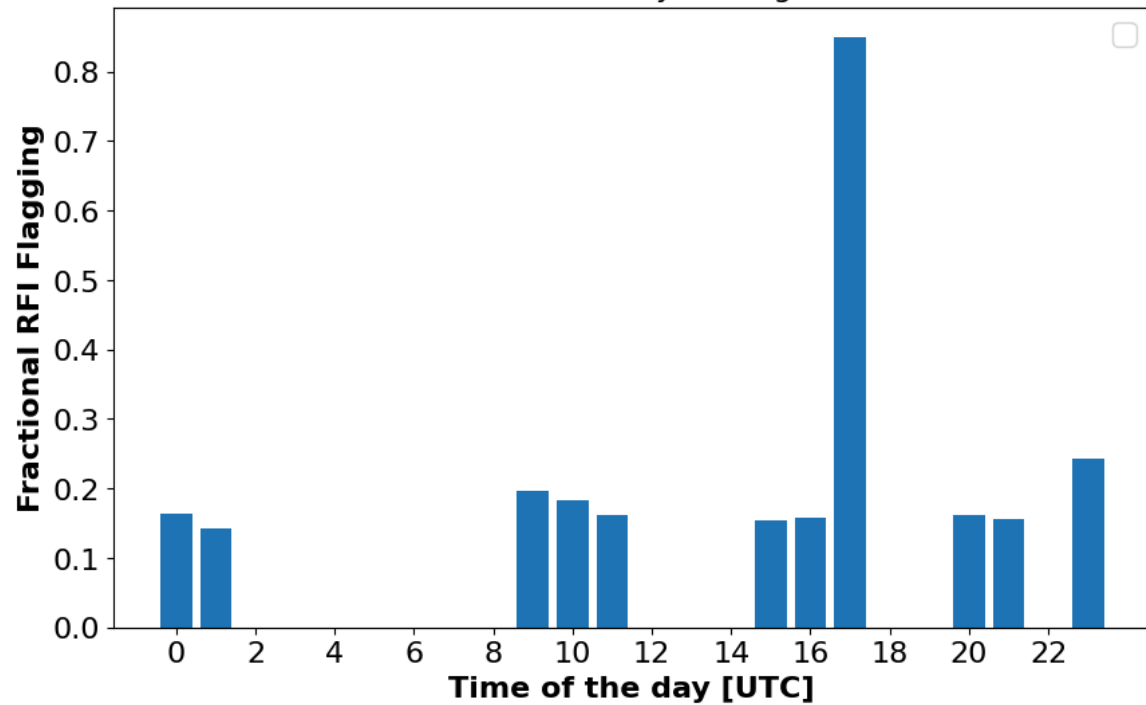


Figure 13: The plot above shows the fractional RFI flagging as a function of time of the day in UTC for August 2024 UHF horizontal cross pol observations.

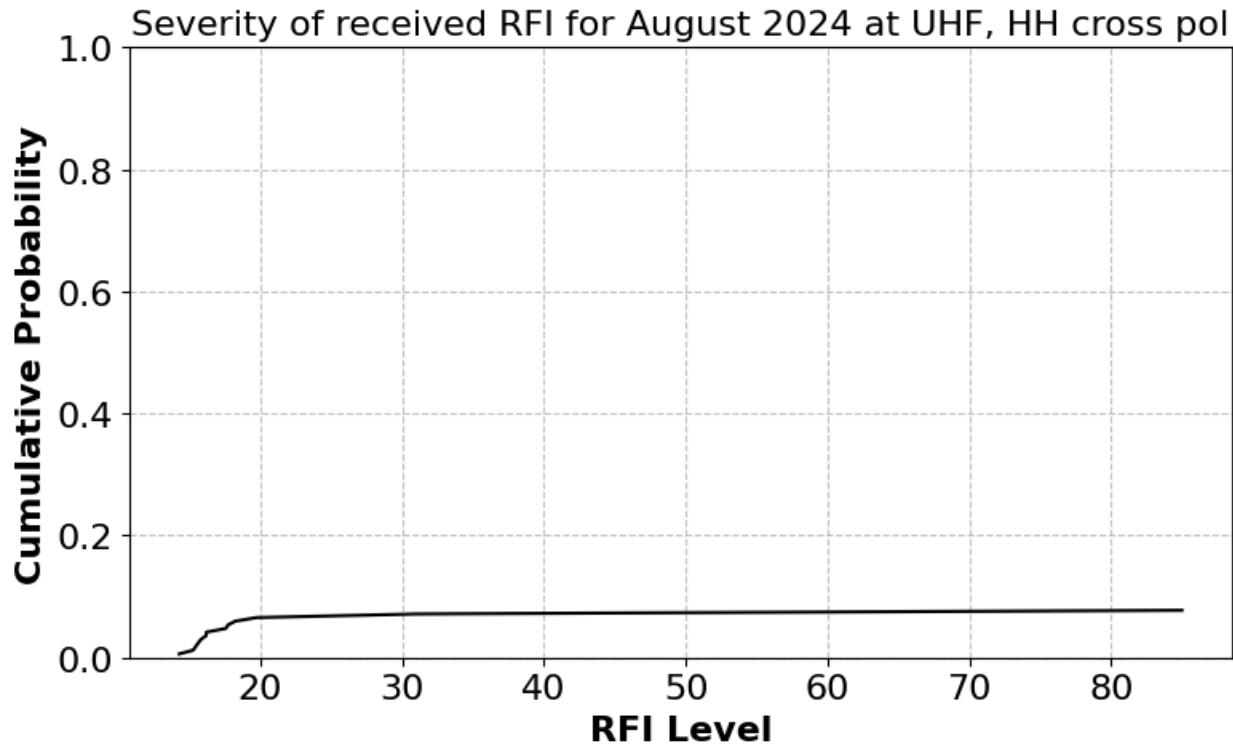


Figure 14: RFI distribution : cumulative probability of RFI as a function of RFI levels.

Figure 13 illustrates the hourly behavior of RFI for the month of August 2024. This view allows us to identify specific time periods affected by RFI. The probability of RFI is relatively low most of the time, until 16:00 UTC, the fractional RFI rises sharply to a maximum of 85%, this suggests a period when RFI is especially prevalent. This could be attributed to a number of causes: This could be due to specific construction activities on site , also could be due to a source under observation at the time. To be able to characterize, we need to look at the frequency spectra at that time of the time.

Figure 14, shows the Empirical Cumulative Distribution Function (ECDF) of RFI levels for August 2024. The x-axis represents the RFI levels, while the y-axis shows the cumulative probability, which represents the proportion of RFI data points that are less than or equal to each RFI level. This is useful for understanding the overall intensity distribution of the RFI events. For example, Figure 14 shows that at RFI level 20, around 7% of the data falls below this level with a steep slope from lowest probability, indicating that low RFI levels are common or very concentrated in that range. Therefore, we can conclude that around 7% of all RFI data points fall at or below an RFI level of 20. The steep initial rise in the ECDF indicates that a large portion of RFI events occur at these lower intensities, while higher RFI levels are much less frequent, occurring only in a small fraction of the observations. Thus, severe RFI events are rare, but they do exist in the dataset.

Moreover, Figure 13 tells ‘when’ RFI occurs and the amount, while Figure 14 provides a statistical summary of ‘how severe’ RFI is overall. These figures combined give time-dependent view and cumulative probability view, allowing us to understand when RFI spikes occur and how often they reach a certain intensity. Another way to look at this: Let us say, you observe a time window with high RFI from the time series plot (Figure 13), e.g 17:00 UTC, you can reference Figure 14 (ECDF) to understand whether that intensity is rare or typical across the observations in a given month.

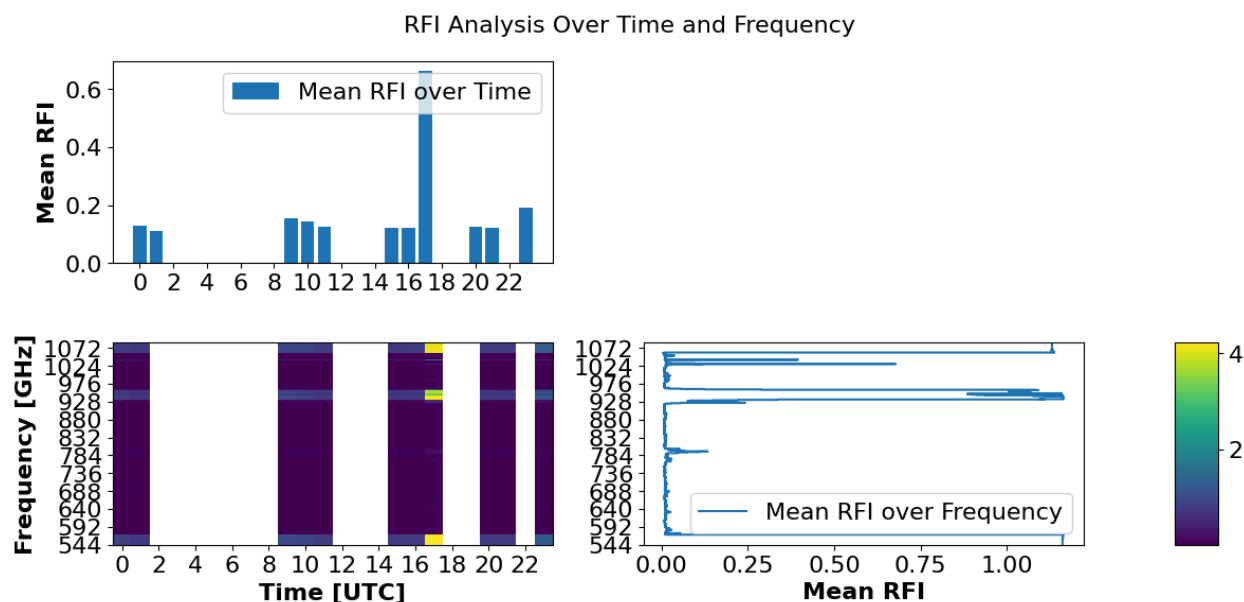


Figure 15 : Mean RFI across time and frequency space

Figure 15, depicts a 2 dimensional view of RFI patterns over time and frequency. The top 1D bar plot shows the mean RFI levels over time exactly the same as Figure 13. The 2D heatmap in the centre represents the detailed RFI activity across time and frequency, where white areas suggest missing data or non-monitored frequency intervals. Notably, intense RFI zones appear as bright yellow or green patches as indicated by the colour bar (far right), notably between 544-592, 800–900 MHz and at higher frequencies above 950 MHz. The 1D plot on the right highlights mean RFI levels over different frequencies, with spikes indicating strong interference at specific frequencies. Fortunately we know the RFI present in these frequencies, GSM downlink – mobile communications. There is a weak spike around 784 MHz which is due to pilling.

References

[1] Sihlangu, I., Oozeer, N. and Bassett, B., 2022. Nature and Evolution of UHF and L-band Radio Frequency Interference at the MeerKAT Radio Telescope. *arXiv:2211.08879*.